

Experiment -

Study of LAN Topology, Network Devices, and Address Configuration.

Problem Statement -

2. Draw the Network topology:-

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point, and end devices (PCs, laptops, etc).

Ans:-

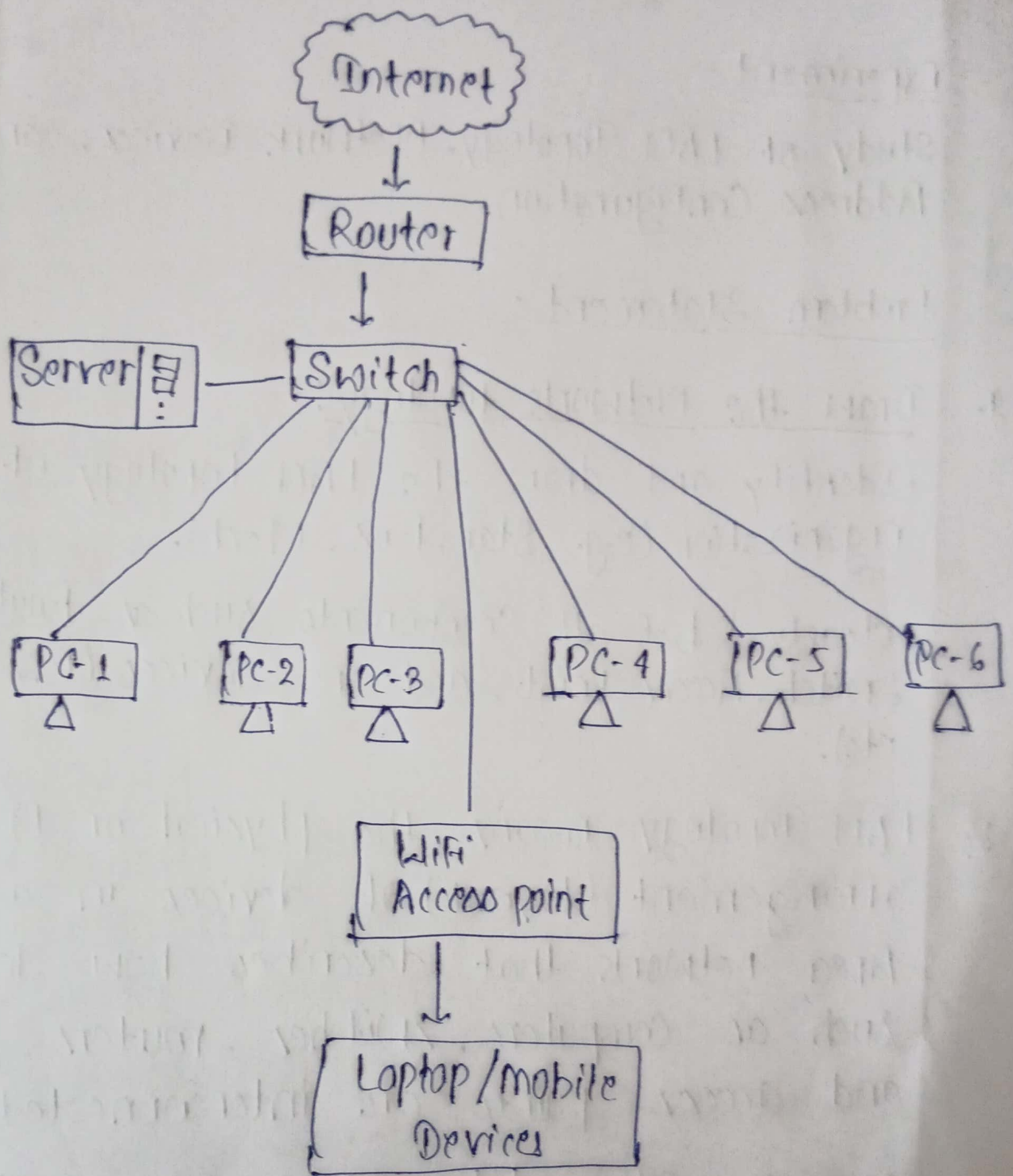
LAN Topology means the physical or logical arrangement (layout) of devices in a local Area network that describes how devices such as Computers, Switches, routers, printers, and access points are interconnected.

Example- College labs mainly uses a Star Topology.

- Star topology - All devices are connected to a central switch or hub.

Now,

Diagram of network topology -



2. Identify Network Devices -

- List all networking devices used in the LANs.
- Mention the function of each device.

Ans. There are following networking devices used in LAN :-

- i) Router
- ii) Switch
- iii) Access point
- iv) Server
- v) Desktop Computers
- vi) Laptop / mobile devices

Functions of listed devices :-

i) Router -

- Connects the College LAN to the Internet and routes data between different networks.

ii) Switch -

- Connects multiple Computers and devices within the LAN and forwards data to the correct destination.

iii) Access point -

- Provides wireless (Wi-Fi) connectivity to laptops and mobile devices.

iv) Server -

- Stores files, applications, and user data, and provides network services to clients.

v) Desktop Computers:-

- End devices used by students and faculty for practical work and accessing network resources.

vi) Laptop/Mobile devices -

- Wireless client devices that access the network through the Access point.

Q.1) Analyze Connection Types:-

- Specify whether connections are:-
 - Wired (Ethernet)
 - Wireless (Wi-Fi)
- mention where each type is used.

Ans. - Connection Types -

1. Wired Connection (Ethernet)

- Medium - Ethernet (LAN) Cable
- Used for - Desktop computers, server, network switch and router.
- Reason - Provides high speed, stable connection, and low latency.

2. Wireless Connection (Wi-Fi)

- medium - Wi-Fi
- Used for - Laptops, smartphones, and other wireless devices through the Access point.
- Reason - Provides mobility and allows users to connect without cables.

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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\DELL> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : mshome.net
    Link-local IPv6 Address . . . . . : fe80::f9f5:16d4:c0c2:c132%10
    IPv4 Address. . . . . : 10.85.23.124
    Subnet Mask . . . . . : 255.255.248.0
    Default Gateway . . . . . : 10.85.16.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:
```

29°C
Mostly cloudy



Search



```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . . : mshome.net
Link-local IPv6 Address . . . . . : fe80::f9f5:16d4:c0c2:c132%10
IPv4 Address. . . . . : 10.85.23.124
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
```

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

```
Connection-specific DNS Suffix . . :
Link-local IPv6 Address . . . . . : fe80::3ee3:3eaa:bdd6:31b3%44
IPv4 Address. . . . . : 172.25.64.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
```

PS C:\Users\DELL> |

Ques 4) Measure Network Speed:

• Find the link speed of:

- Wired network (if available)

- Wireless network

• Mention the tools / Commands used.

Ans.

Tool Used -

• Windows Task manager

Command Used -

• none

① Wired network (not available)

② Wireless network

• measured speed - 72.0 Kbps

Performance

Run new task



CPU
1% 1.86 GHz



Memory
7.4/15.7 GB (47%)



Disk 0 (C: D:)
SSD (NVMe)
0%



Wi-Fi
Wi-Fi
S: 0 R: 72.0 Kbps

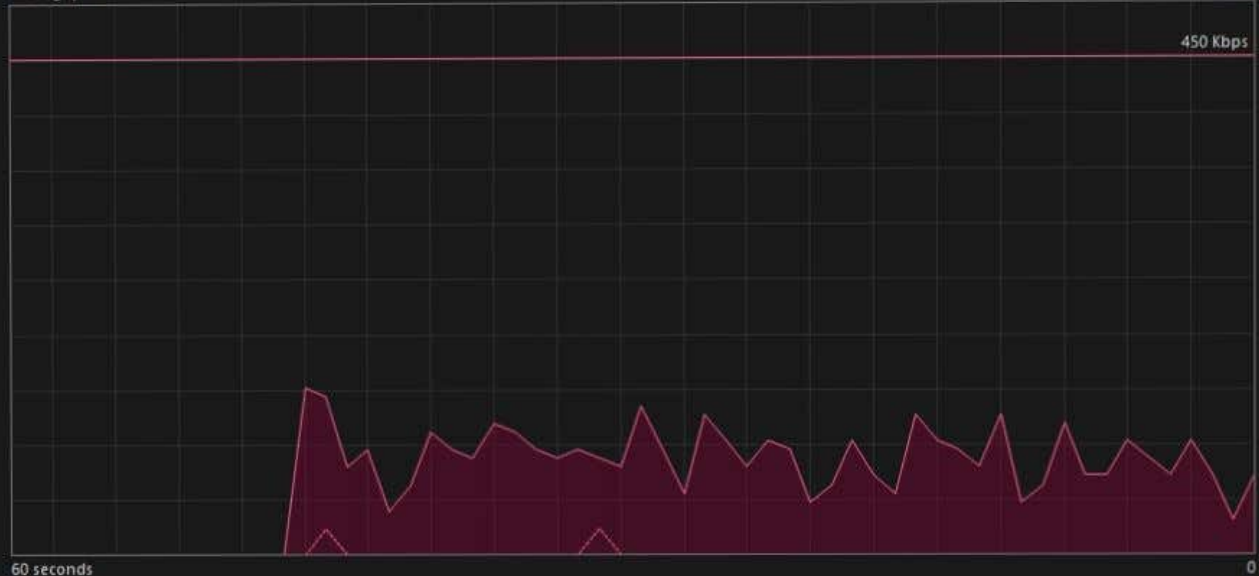


GPU 0
Intel(R) Iris(R) Xe G...
0%

Wi-Fi

Intel(R) Wi-Fi 6 AX201 160MHz

Throughput



Send	Adapter name:	Wi-Fi
0 Kbps	SSID:	GECK-Campus-WiFi
	DNS name:	mshome.net
Receive	Connection type:	802.11ac
72.0 Kbps	IPv4 address:	10.85.23.124
	IPv6 address:	fe80::f9f5:16d4:c0c2:c132%10
	Signal strength:	



Search



ENG
IN



9:06 PM
7/18/2026

Q.5) Find System Network Details:-

- Determine and record the following of your Computer:-

- MAC Address
- IP Address (IPv4)
- Subnet Mask

- Mention the Command(s) used.

Ans - Command used - `ipconfig /all`

- MAC Address - 90-65-84-93-E7-4D
- IP Address - 10.85.23.124
- Subnet Mask - 255.255.248.0

```
Windows PowerShell X + v

PS C:\Users\DELL> ipconfig /all

Windows IP Configuration

Host Name . . . . . : SinghSahiba
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mshome.net


Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 90-65-84-93-E7-4E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes


Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 92-65-84-93-E7-4D
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes


Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : mshome.net
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : 90-65-84-93-E7-4D
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 29°C
Mostly cloudy



mshome.net

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : 90-65-84-93-E7-51
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek PCIe GbE Family Controller
Physical Address. : 4C-D7-17-62-F9-FA
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Description : Hyper-V Virtual Ethernet Adapter
Physical Address. : 00-15-5D-6C-D9-F7
DHCP Enabled. : No
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::3ee3:3eaa:bdd6:31b3%44(Preferred)
IPv4 Address. : 172.25.64.1(Preferred)
Subnet Mask : 255.255.240.0
Default Gateway :
DHCPv6 IAID : 738202973
DHCPv6 Client DUID. : 00-01-00-01-2D-9C-96-AC-4C-D7-17-62-F9-FA
NetBIOS over Tcpip. : Enabled

PS C:\Users\DELL> |

Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. : 92-65-84-93-E7-4D
DHCP Enabled. : No
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : mshome.net
Description : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. : 90-65-84-93-E7-4D
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::f9f5:16d4:c0c2:c132%10(Preferred)
IPv4 Address. : 10.85.23.124(Preferred)
Subnet Mask : 255.255.248.0
Lease Obtained. : Saturday, July 18, 2026 7:26:47 PM
Lease Expires : Sunday, July 19, 2026 7:56:28 AM
Default Gateway : 10.85.16.1
DHCP Server : 10.85.16.1
DHCPv6 IAID : 93349252
DHCPv6 Client DUID. : 00-01-00-01-2D-9C-96-AC-4C-D7-17-62-F9-FA
DNS Servers : fe80::deea:e7ff:feb2:dccf%10
 8.8.8.8
 1.1.1.1
NetBIOS over Tcpi. : Enabled
Connection-specific DNS Suffix Search List :
 mshome.net

Q 6) Explain Observations:-

• Briefly explain:-

- Difference b/w wired and wireless speed.

- Role of subnet mask in networking.

Ans. There are following differences:-

1. Wired Network (Ethernet):-

• A wired network transfers data through Ethernet cables.

• It provides higher speed and more stable connectivity.

• It has lower latency and fewer chances of interferences.

• It is commonly used in Computer labs, offices and servers.

2. Wireless Network (Wi-Fi):-

• A wireless network transfers data using radio waves.

• Its speed depends on signal strength, distance, and interference.

• It provides mobility and allows devices to connect without cables.

Role of Subnet Mask in Networking -

A Subnet Mask is used to divide an IP address into two parts:-

- i) Network Port
- ii) Host Port

- It identifies the network and host portion of an IP address.
- It helps devices communicate within the correct network.
- It helps routers determine where to send data packets.

Ex- IP Address : 172.25.71.217

Subnet Mask : 255.255.240.0

- It helps identify the network to which the device belongs.